

LAYER: InstructPix2Pix Identity-Layer Geometry Results

Stage-B targeted layer optimization with Stage-A identity-layer context

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Objective: $Z = 1 - \text{cosine_similarity}(\text{pool}(\text{layer}(\text{original})), \text{pool}(\text{layer}(\text{perturbed})))$. Loss: $\text{loss} = -Z$. No visual counter-loss was used.

Run matrix

layers	prompts	images	runs	iterations
1.0000	2.0000	4.0000	8.0000	400

Aggregate results

layer	prompt	runs	mean dZ	max dZ	mean SSIM	min SSIM	invalid
UNET.conv_in	add black sunglasses	4.0000	0.0005332	0.002133	0.9972	0.9924	0
UNET.conv_in	add headphones	4.0000	0.0005209	0.002084	0.9948	0.9831	0

Per-run final values

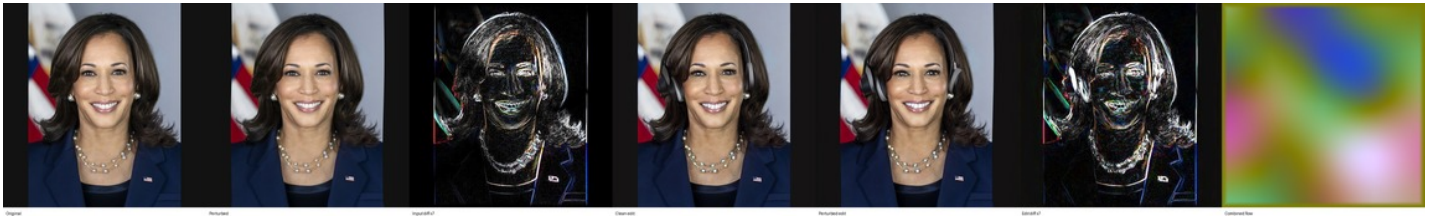
validity	layer	prompt	image	source	dZ	SSIM	out SSIM	max disp
valid_or_visually_check	UNET.conv_in	add black sunglasses	kamala_harris_01	best_valid_checkpoint	0.002133	0.9924	0.9916	2.4273
valid_or_visually_check	UNET.conv_in	add headphones	kamala_harris_01	best_valid_checkpoint	0.002084	0.9831	0.9775	4.4882
valid_or_visually_check	UNET.conv_in	add black sunglasses	barack_obama_01	best_valid_checkpoint	0	0.9987	0.9974	0.5777
valid_or_visually_check	UNET.conv_in	add black sunglasses	joe_biden_01	best_valid_checkpoint	0	0.999	0.9939	0.5777
valid_or_visually_check	UNET.conv_in	add black sunglasses	michelle_obama_01	best_valid_checkpoint	0	0.9987	0.9976	0.5777
valid_or_visually_check	UNET.conv_in	add headphones	barack_obama_01	best_valid_checkpoint	0	0.9987	0.9957	0.5777
valid_or_visually_check	UNET.conv_in	add headphones	joe_biden_01	best_valid_checkpoint	0	0.999	0.9884	0.5777
valid_or_visually_check	UNET.conv_in	add headphones	michelle_obama_01	best_valid_checkpoint	0	0.9987	0.9943	0.5777

Image strips

unet.conv_in / add black sunglasses / kamala_harris_01 - valid_or_visually_check



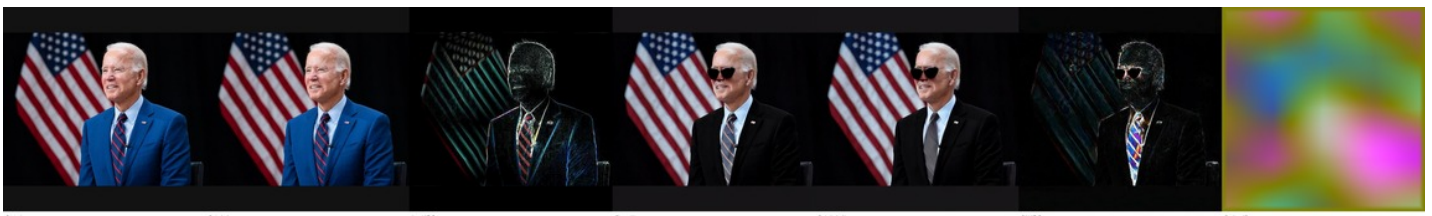
unet.conv_in / add headphones / kamala_harris_01 - valid_or_visually_check



unet.conv_in / add black sunglasses / barack_obama_01 - valid_or_visually_check



unet.conv_in / add black sunglasses / joe_biden_01 - valid_or_visually_check



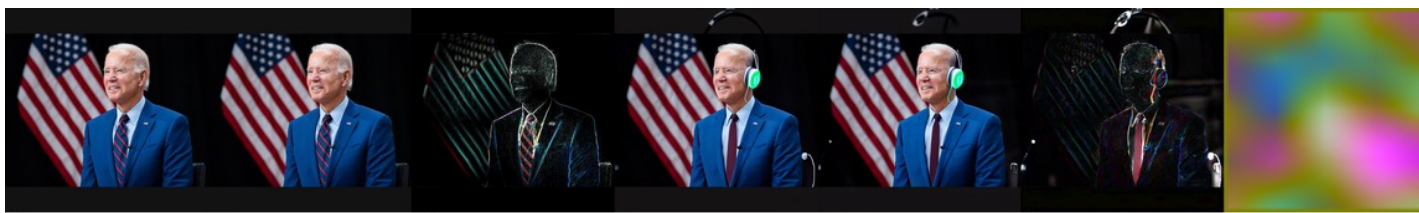
unet.conv_in / add black sunglasses / michelle_obama_01 - valid_or_visually_check



unet.conv_in / add headphones / barack_obama_01 - valid_or_visually_check



unet.conv_in / add headphones / joe_biden_01 - valid_or_visually_check



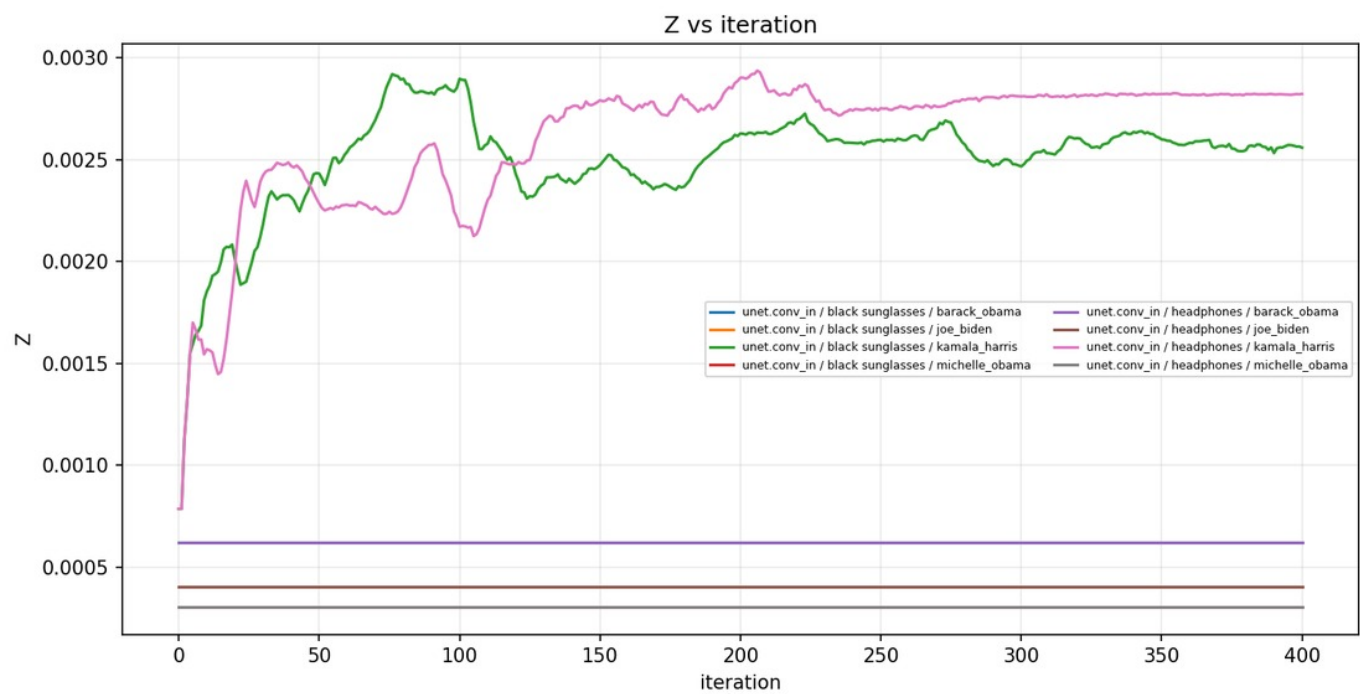
unet.conv_in / add headphones / michelle_obama_01 - valid_or_visually_check



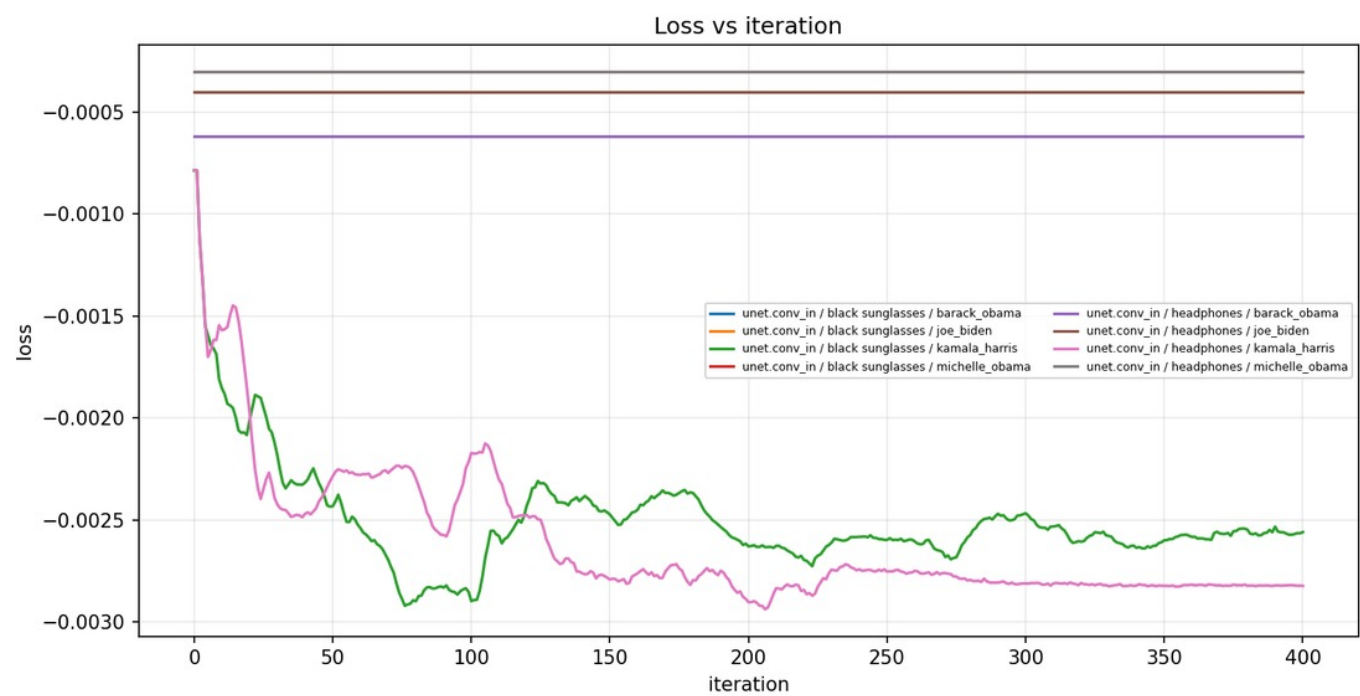
Graphs

Cross-run graphs

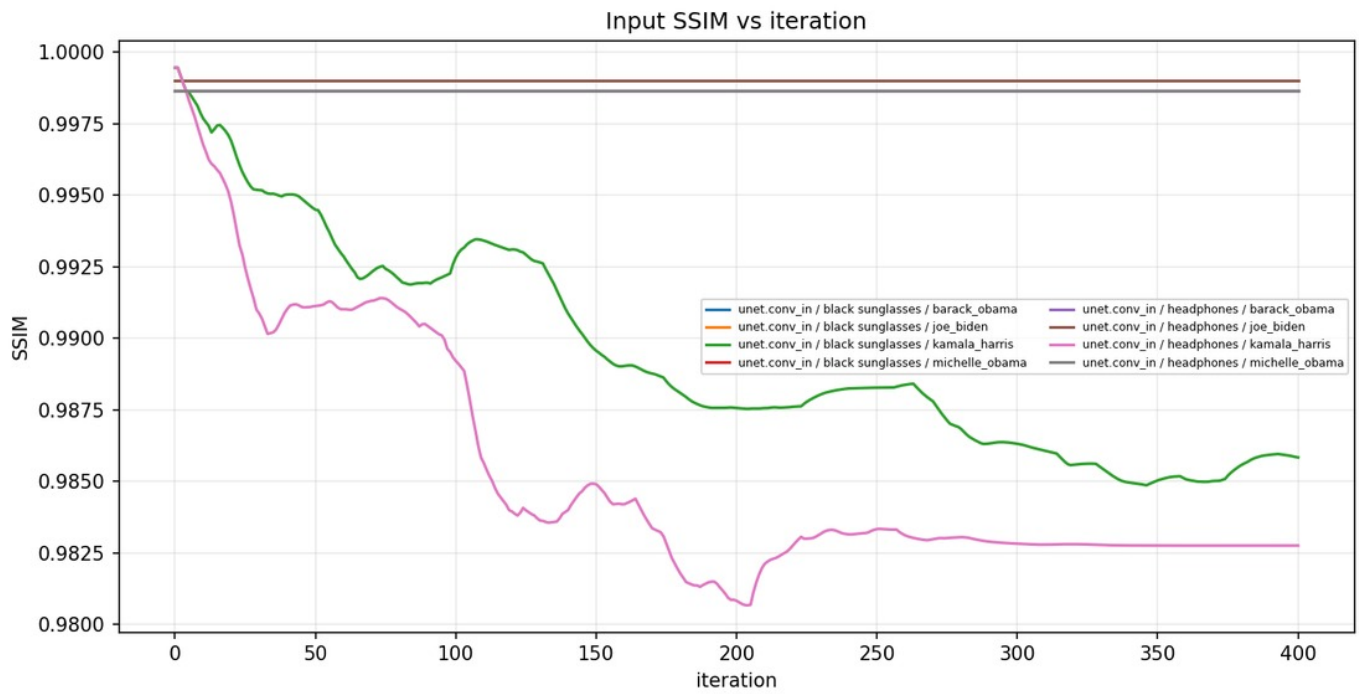
Z vs iteration



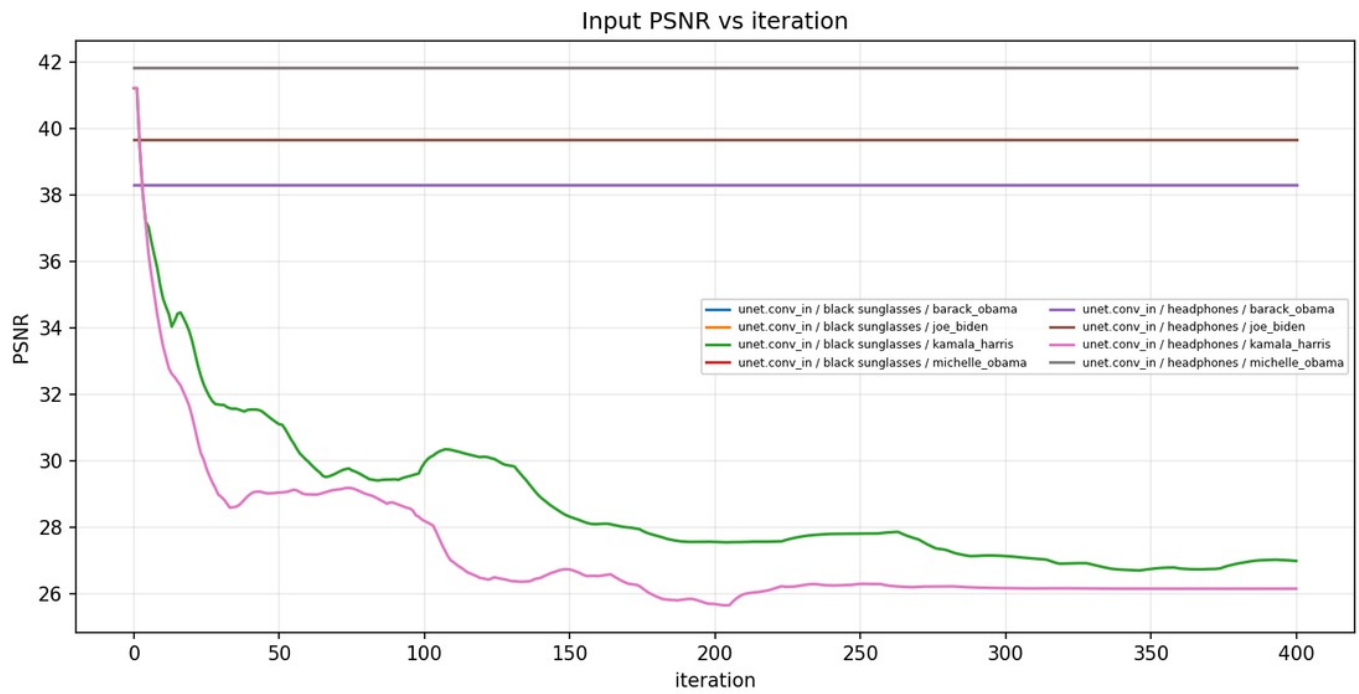
Loss vs iteration



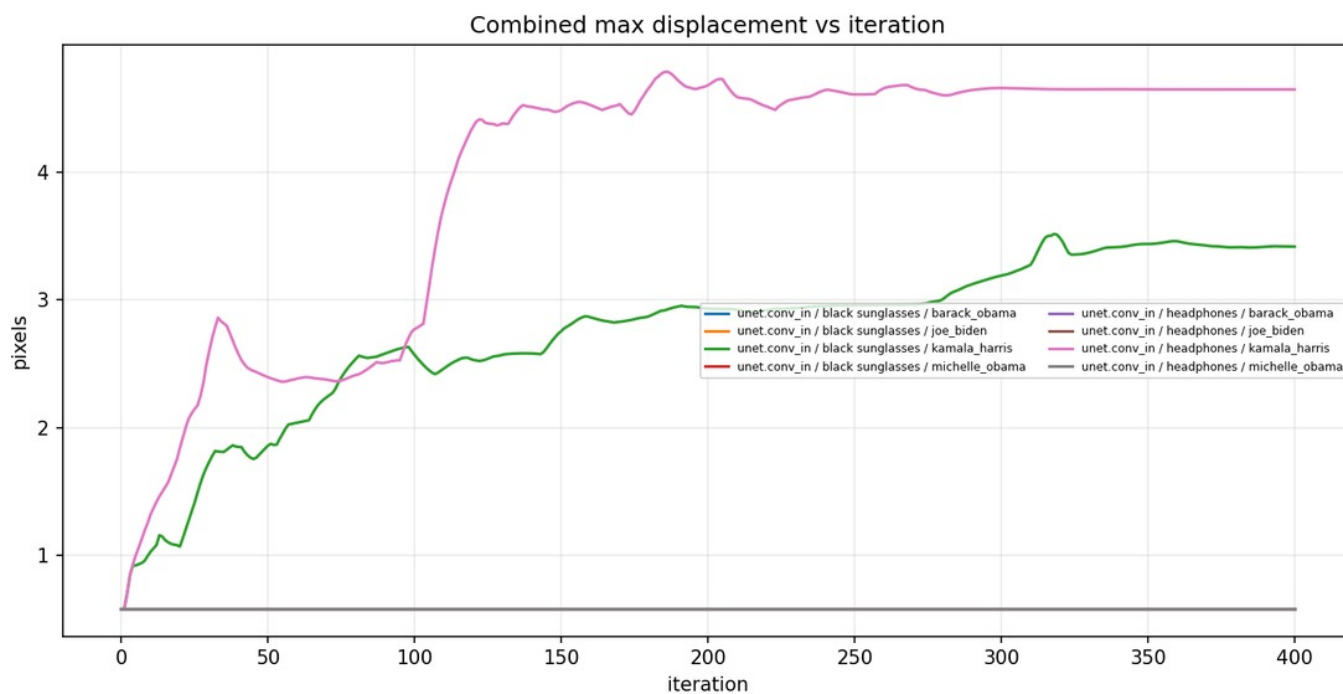
Input SSIM vs iteration



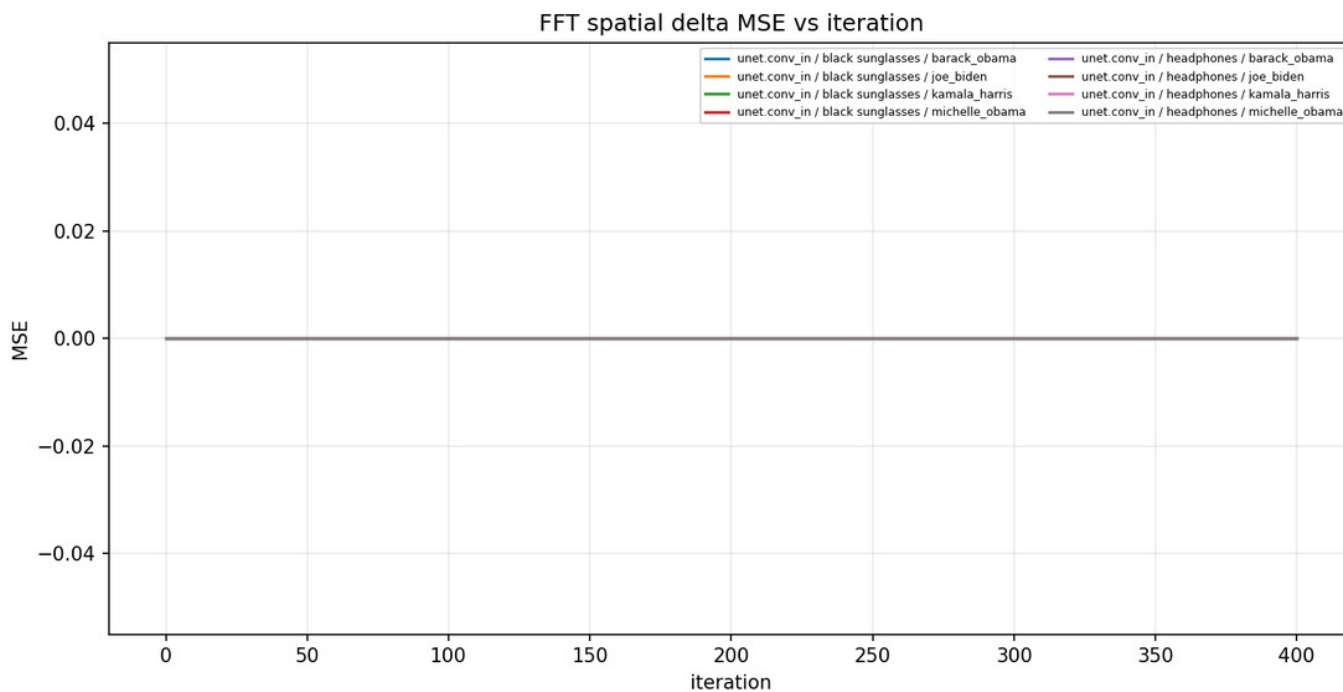
Input PSNR vs iteration



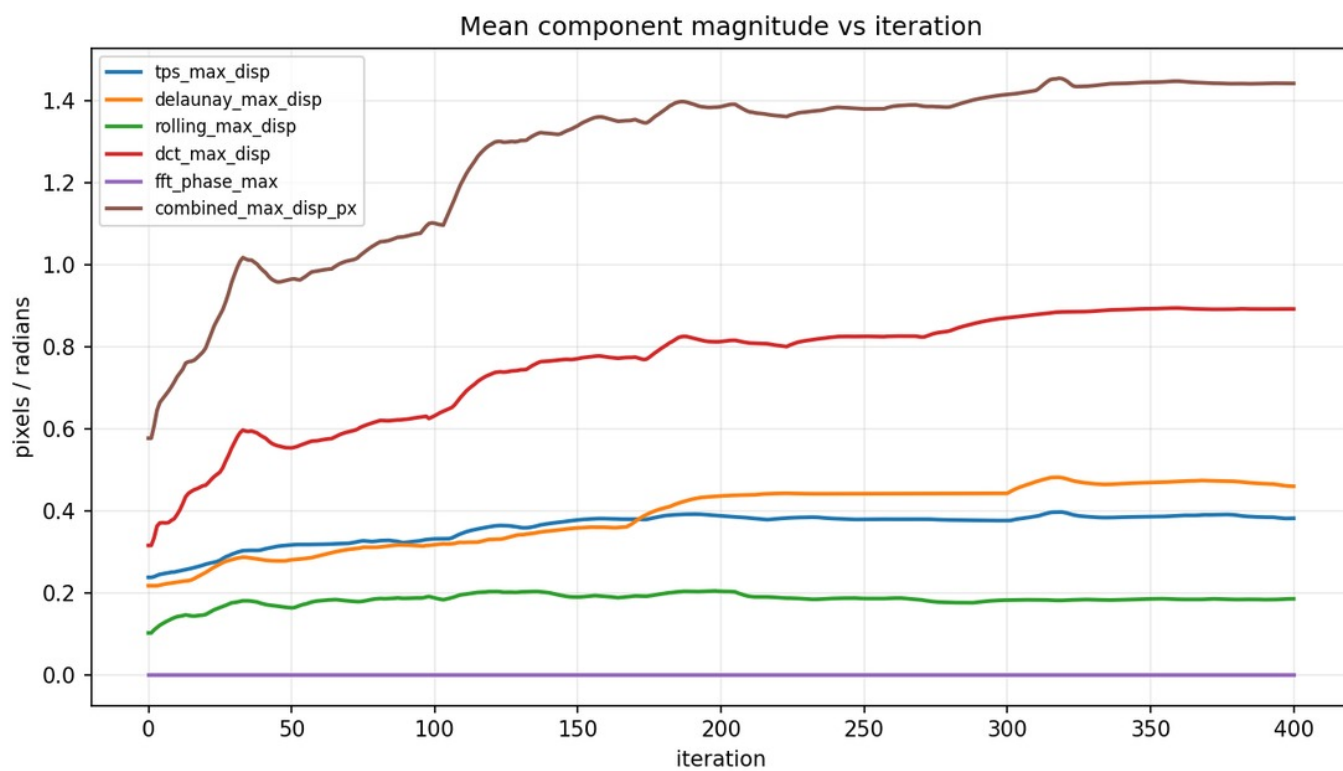
Combined max displacement vs iteration



FFT spatial delta MSE vs iteration



Component magnitude vs iteration



Z increase vs input SSIM

